

(Part A) Project Setup and Understanding

Project Topic

This project analyzes billing patterns using the 2024 *Centers for Medicare and Medicaid Services*

(CMS) *Medicare Physician and Other Practitioners - by Provider and Service* public dataset (about 9.7M rows, 28 features), a record of services, charges, and payments aggregated by provider and procedure. This project uses unsupervised anomaly detection on this dataset and outlines any results in a Power BI dashboard.

Problem Statement

Insurance and public health programs processing high volume datasets may not be able to manually review information with equal attention, contributing to analyst fatigue over time. Without a verified fraud label in the dataset, supervised classification isn't possible. Analysts must find a way to focus limited review capacity on abnormal billing behavior with respect to similar providers and services without implying fraudulent or lesser services. A useful analytics system would streamline this process, but preserve an analyst's ability to make the final determination.

Objective

Make a reproducible pipeline assigning each provider-service billing record a 0-100 anomaly risk score, where higher values indicate more unusual peer-relative behavior. The project will:

1. compare Isolation Forest against z-score and percentile baselines, among other unsupervised methods to generate the anomaly risk ranking,
2. test ranking stability with one or more transparent baselines,
3. present results in a Power BI dashboard displaying what was flagged, where, and why, and
4. effectively communicate its limitations so that a flag is not misrepresented as fraud.

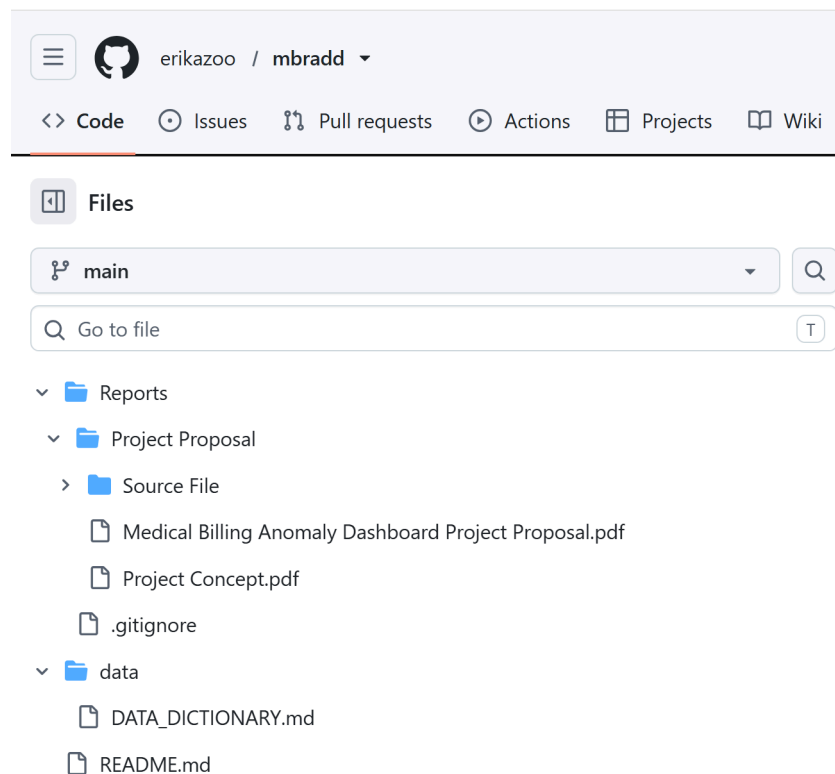
Data Science Questions

1. Which Medicare provider-service billing patterns are most unusual relative to comparable peers?
2. Which features most frequently drive high-risk rankings, such as service intensity, submitted-charge ratios, or unusual procedure concentration?
3. How stable are the highest-risk rankings when model assumptions, contamination levels, or feature sets change?
4. How much does the top-risk set change when random seeds, model settings, contamination assumptions, or feature sets change?

5. Do Isolation Forest and transparent percentile/z-score methods identify overlapping high-risk patterns?
6. What billing characteristics distinguish the top 1%, 5%, and 10% anomaly-risk groups from the overall dataset?
7. For selected high-risk providers, can the dashboard show a clear and reproducible explanation for the score?

Github Repository Organization

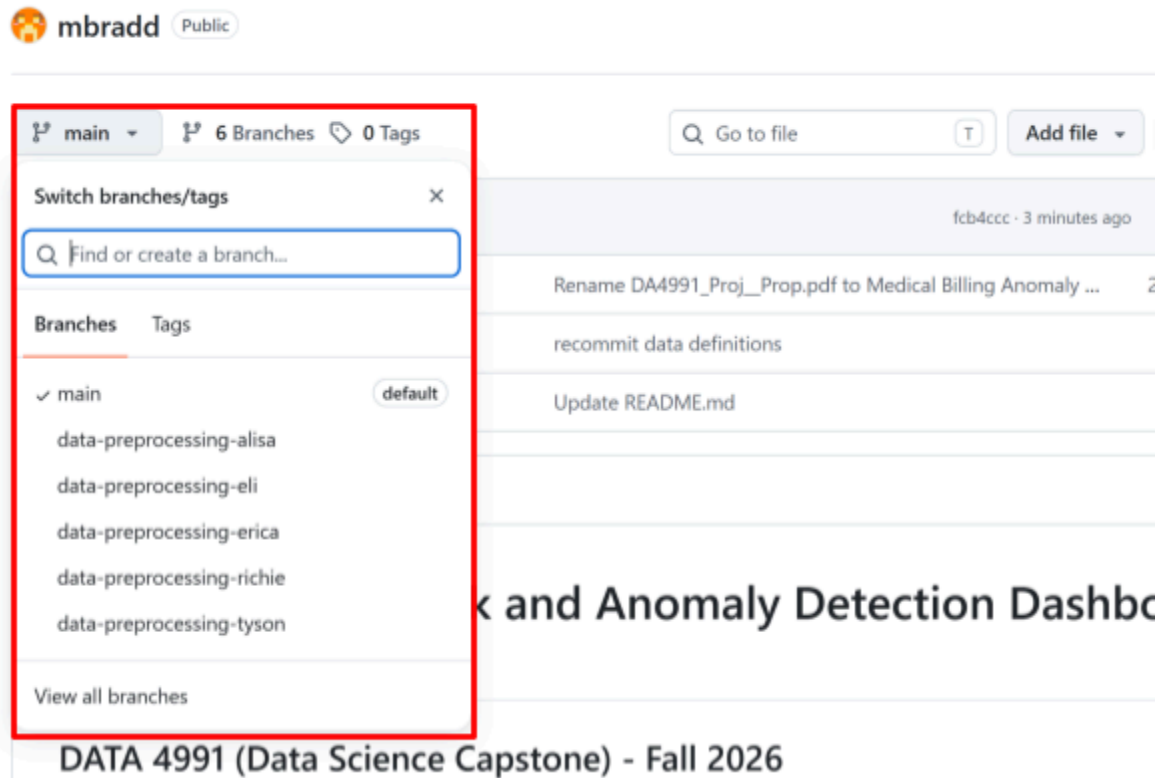
Initial documentation, such as reports and dataset descriptions, are displayed in the main repository. It will be structured as a branch-per-developer workflow merged into main. Code and documentation from each phase will be grouped and merged into the main repository as the workflow progresses, then organized into folders as needed. Currently, the main repository has a folder for reports and data, along with a README.



What represents the repository structurally the best can be visualized via *data-preprocessing-erica*, with data preprocessing work, Git file documentation, and all project reports so far. More data visualization and charts can be found in other branches.

All necessary visualization needs to be placed within that branch, with other primary documentation within other branches documented in Part B of this project. Branch

data-preprocessing-erica will be merged into main, and redistributed for team members working on the next part of the project. To track the live progress per developer of this project on GitHub prior to the completion of a sprint, refer to each branch by clicking on the branch menu as shown:



Dataset and Resources

The project uses the 2024 *Medicare Physician & Other Practitioners - by Provider and Service* dataset (9.7M rows, 28 columns), a subset of the 2013-2024 Centers for Medicare and Medicaid Services datasets. It is updated annually. Source link:

<https://data.cms.gov/provider-summary-by-type-of-service/medicare-physician-other-practitioner/s/medicare-physician-other-practitioners-by-provider-and-service/data>

Of the 28 columns, 20 features were selected for likely use in overall analysis: ['Rndrng_NPI', 'Rndrng_Privr_Last_Org_Name', 'Rndrng_Privr_First_Name', 'Rndrng_Privr_Crdntls', 'Rndrng_Privr_Ent_Cd', 'Rndrng_Privr_City', 'Rndrng_Privr_State_Abrvtn', 'Rndrng_Privr_Type', 'Rndrng_Privr_Mdcr_Prtcptg_Ind', 'HCPCS_Cd', 'HCPCS_Desc', 'HCPCS_Drug_Ind', 'Place_Of_Srv', 'Tot_Benes', 'Tot_Srvcs', 'Tot_Bene_Day_Srvcs', 'Avg_Sbmtld_Chrg', 'Avg_Mdcr_Alowd_Amt', 'Avg_Mdcr_Pymt_Amt', 'Avg_Mdcr_Stdzd_Amt']. A smaller subset of costs, provider ID, and location is being selected for the clustering algorithms.

Project members will choose their tools and be responsible for its visibility (.ipynb) on the GitHub repository. Because the data exceeds the GitHub storage limits, one may follow the *data-preprocessing-erica* branch to be able to download the dataset from the URL on a marimo python notebook, managed via VSCode in 'data_preprocessing_ez.ipynb'.

Microsoft Power BI (Power Business Intelligence) shall host the dashboard for this project, as it is a robust and interactive software with built-in tools such as Power Query to connect our data and DAX for KPI cards and measures.

Proposed Workflow

This project follows the agile methodology best, with members iterating on their own sprints roughly at the same time and adjusting as requirements evolve. The workflow can be visualized with the backlog table below:

Phase	Step	Process	Output
Prepare	Dataset	Download CMS Public Dataset	Raw data
	Data Preprocessing	Check and explain missingness, duplicates, odd ranges. Determine relevant columns.	Clean table
	Feature Engineering	Select model features, modifying categorical data for model use or evaluating correlation.	Feature table
	Data Analysis	Visualize distributions and relationships. Use them to explain the dataset and what needs to happen next.	Charts/Graphs
Detect	Statistical Baseline	Z-Scoring, Percentile flags	Transparent flags
	Isolation Forest	Unsupervised model isolating unusual records, with a potential backup methodology depending on efficacy	Anomaly scoring/ranking
Validate	Model Evaluation, Stability Testing	Rerun seeds and settings, compare Isolation Forest to baseline	Trustworthiness of rankings
Deliver	Anomaly Scores and Feature Explanations	0-100 risk score and record relevant features	Scored table and reasonings
	BI Dashboard	Power BI pages: Summary, Review Queue, Provider In-Depth, Peer analysis, Model	Interactive dashboard

		Transparency	
	Distribution	Demo analyst review using the BI deliverable, summary of findings, complete Github Repository for reproduction and evolution	Cohesive presentation

Current Progress and Project Setup

Preprocessing of the data has been completed and verified by all team members, progress viewable in Part B. Feature selection is an iterative process, as additional statistical analysis is currently being conducted for effective inputs to Isolation Forest.

We are still trying to decide which inputs best separate unusual billing behavior, including submitted-charge-to-payment ratios and service intensity. The selected features will feed the Isolation Forest (and DBSCAN as a possible backup method).